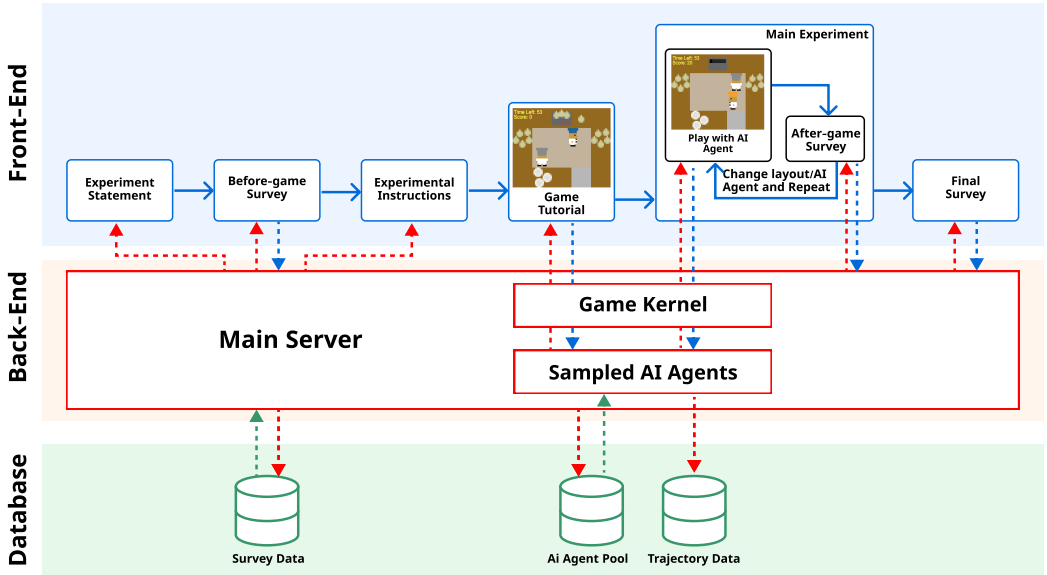

Figure 6: This figure outlines the structure of our human-AI experimental platform, including front-end, back-end, and database components. The front-end, where participants interact, starts with a pre-game process (experiment statement, before-game survey, and experimental instructions) and moves into game stages involving a game tutorial and main experiment with multiple rounds against varying AI agents. A survey follows each round, and a final survey concludes the experiment. All data, including player paths and AI weights, are stored in the database.

settings, world objects, and game controls through both text and images. Following the instruction page, human players need to participate in a trial game to further practice how to play the game. Both the instruction page and trial game contribute to participants gaining a preliminary understanding of the experiment.

Gaming and Questionnaire Pages. The next component is the core of the Human-AI experiment, where participants play with randomly sampled AI teammates and complete questionnaires to evaluate the performance of these AI partners. Specifically, in our initial platform setup (more details are provided in Section 7.6.1), participants play with two different AI teammates on five Overcooked layouts, resulting in 10 games per participant. After each game, they are required to complete a questionnaire that assesses their cooperative gameplay with the AI partner. The final questionnaire primarily focuses on coordination with the AI agent with whom they collaborated and the game they just played. Upon completing all games, participants are directed to the final questionnaire page, where they are asked to provide preliminary feedback on AI agents based on their overall performance, considering factors such as fluency, legibility, and reliability. In contrast to individual game questionnaires, the purpose of this feedback is to analyze and compare the capacities of different agents.

The Human-AI Experiment Platform is designed for seamless customization, enabling users to easily modify various aspects of the system, such as statements, instructions, questionnaires, and game settings. Most customizations can be achieved simply by editing configuration files, streamlined for users to adapt the platform to their specific needs.

6.3 Experimental Scale

In our effort to thoroughly examine the performance of human-AI coordination in a zero-shot scenario, we advocate for a systemically designed experimental scale. This scale incorporates an expansive range of subjective metrics that appraise individual games involving AI agents while providing a holistic comparison across multiple coordinated players. The metrics utilized extend beyond the superficial, capturing elements such as intention, contribution, and team dynamics during human-AI interactions. Furthermore, the evaluation process encompasses additional aspects such as fluency, participant preferences, and comprehension across all collaborated agents. More specifically, participants are requested to participate in after-game surveys and final-game surveys to assess the performance of AI agent collaboration. Each of these surveys comprises three questions that use a 7-point Likert scale, illustrated in Table 1. The selected survey questions for our experiment draw heavily from a pool of analogous questions found in (Hoffman, 2019). For the after-game surveys, the 7-point Likert scales are arranged with 1 denoting "strongly disagree" and 7 indicating "strongly agree". In the final experiment questionnaire, ratings from 1 to 7 reflect preferences from "strongly favoring the first agent" to "strongly favoring the second agent", for instance, a rating of 4 symbolizes an absence of preference. The rating system adopts a star-assigning format, with the rules clearly stated at the onset of each questionnaire.

Table 1: Evaluation statements for the after-game and final questionnaires consist of three statements each. After-game statements aim to assess the cooperative experience with the AI agent in a single game, while final statements require participants to compare the two AI partners and rate their performance after all games have been completed.

Type	Index	Scale statement
After-game	Q1	The agent and I have good teamwork.
	Q2	The agent is contributing to the success of the team.
	Q3	I understand the agent's intentions.
Final	Q1	Which agent cooperates more fluently?
	Q2	Which agent did you prefer playing with?
	Q3	Which agent did you understand with?

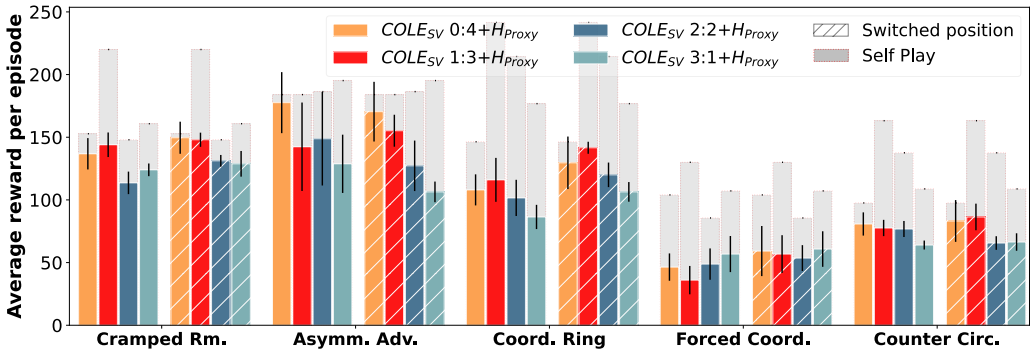

Figure 7: The result of the combining objectives' effectiveness evaluation. Mean episode rewards over 400 timesteps trajectories for COLE_{SV} s with different objective ratios 0:4, 1:3, 2:2, and 3:1, paired with the unseen human proxy partner H_{proxy}. The ratios 0:4, 1:3, 2:2, and 3:1 denote varying proportions between individual and cooperative compatible objectives. The gray bars behind present the rewards of self-play.

7. Experiments

We carry out a series of experiments involving AI agents, human proxies, and human players to assess the performance of COLE_{SV} compared to the baseline methods when collaborating with partners in zero-shot settings. Our experiments focus on the following research questions (RQ):

- RQ1: What is the optimal balance between individual and socially compatible objectives? (Section 7.1)
- RQ2: How do the COLE_{SV} and baseline agents perform when cooperating with agents of varying skill levels in a zero-shot setting? (Section 7.2)
- RQ3: Does COLE_{SV} effectively address the issue of cooperative incompatibility? (Section 7.3)
- RQ4: How do each modules within COLE_{SV} contribute to its overall performance? (Section 7.4)
- RQ5: Which solver exhibits superior performance? (Section 7.5)
- RQ6: How do the COLE_{SV} and baseline agents perform when working with humans in a zero-shot setting, and which algorithm do human users prefer? (Section 7.6)

7.1 Evaluation of Combining Objectives' Effectiveness

We construct evaluations with different ratios between individual and cooperative compatible objectives, such as 0:4, 1:3, 2:2, and 3:1. These studies demonstrate the effectiveness of optimizing both individual and cooperative incompatible goals.